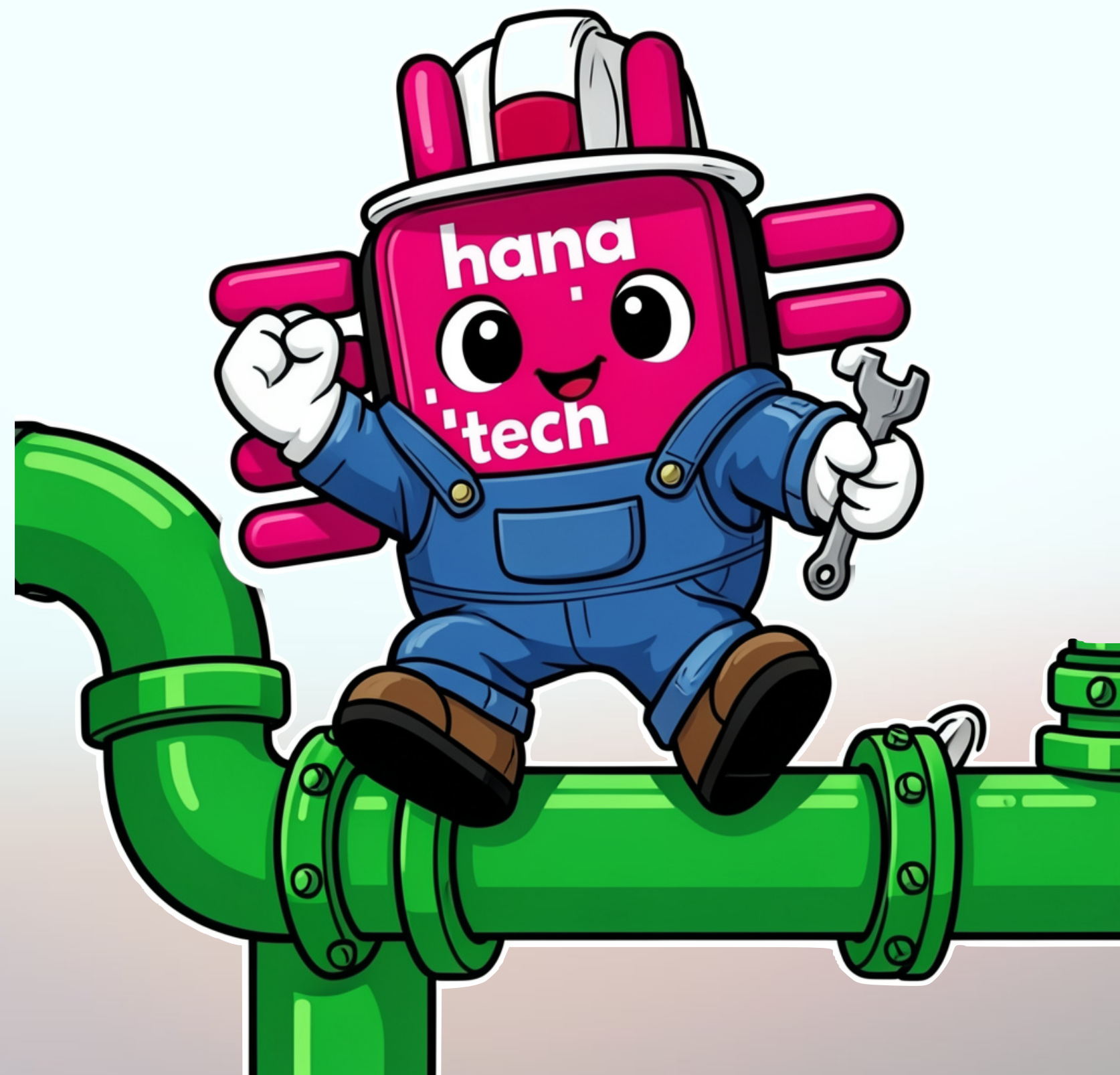


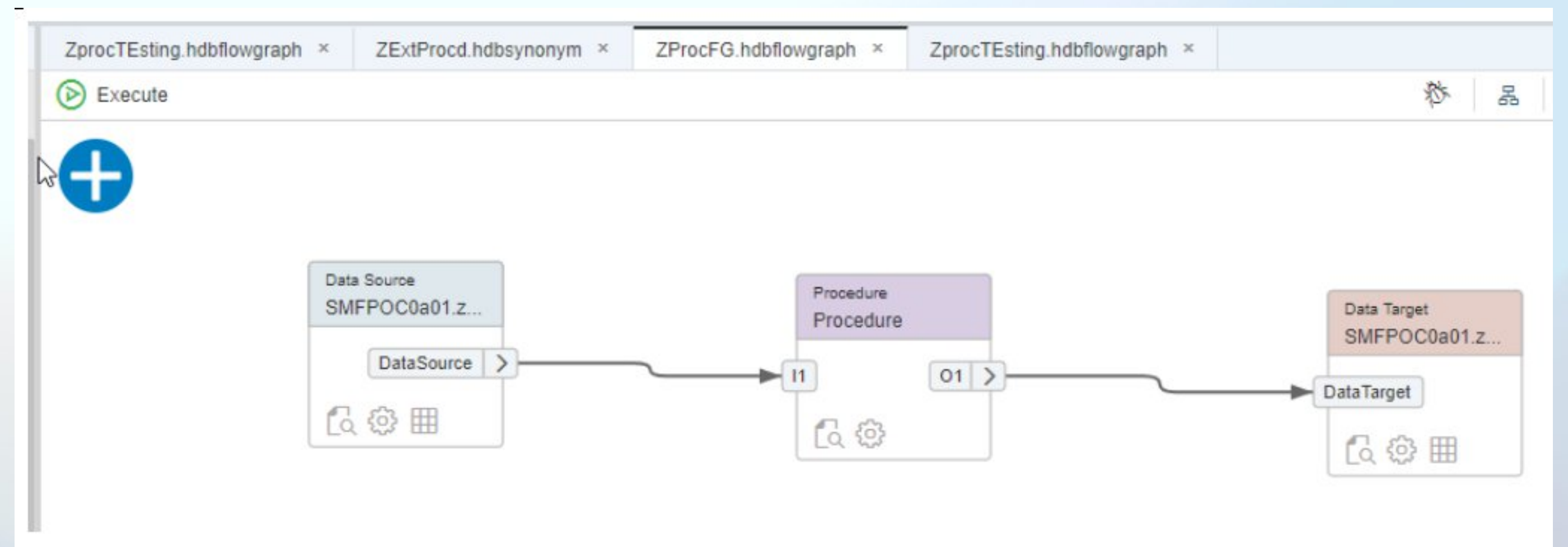
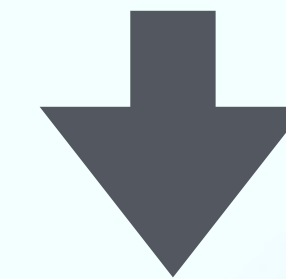
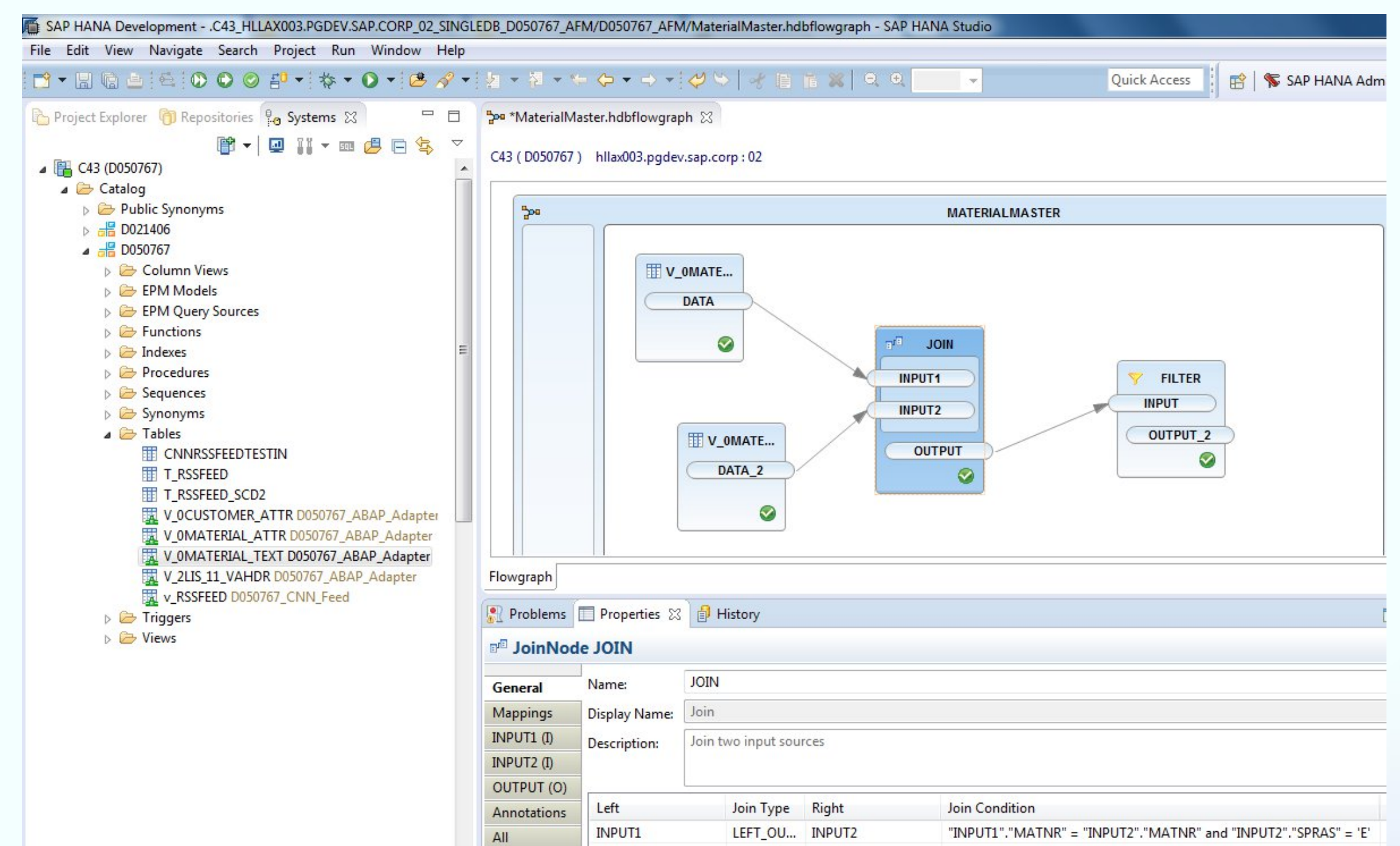
SDI and SDQ Forgotten Knowledge?



Who am I?

Timo Wagner

- Freelance Software Engineer
(Focus on SAP HANA technologies)
- Part of the k-42 e.G. **k-42**
- Former Developer @ SAP
Worked on HANA SDI editors,
DPAgent, Adapters, Agile Data
Preparation and
Data Hub / Data Intelligence



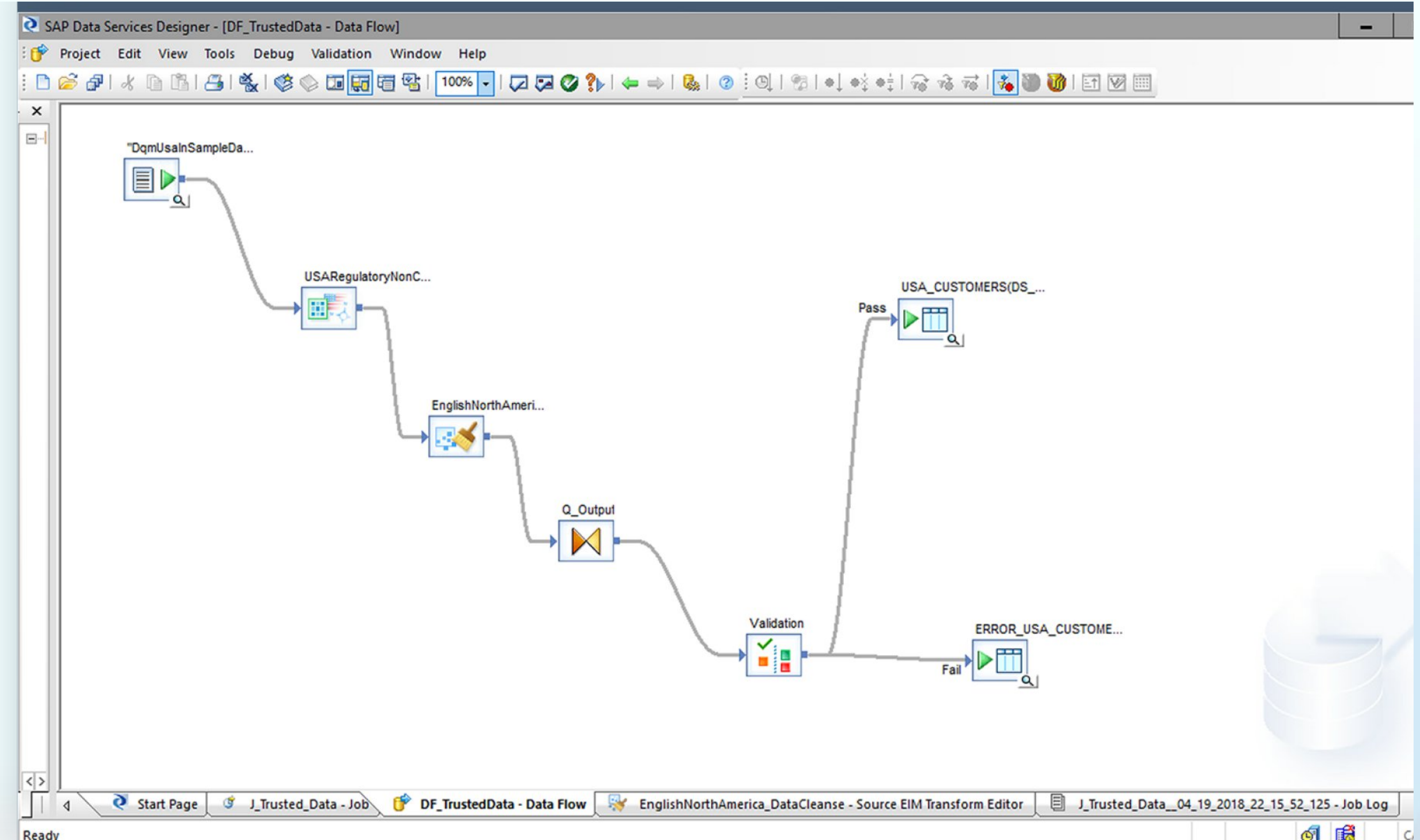
How did it all start?



How did it all start?

SAP Data Services

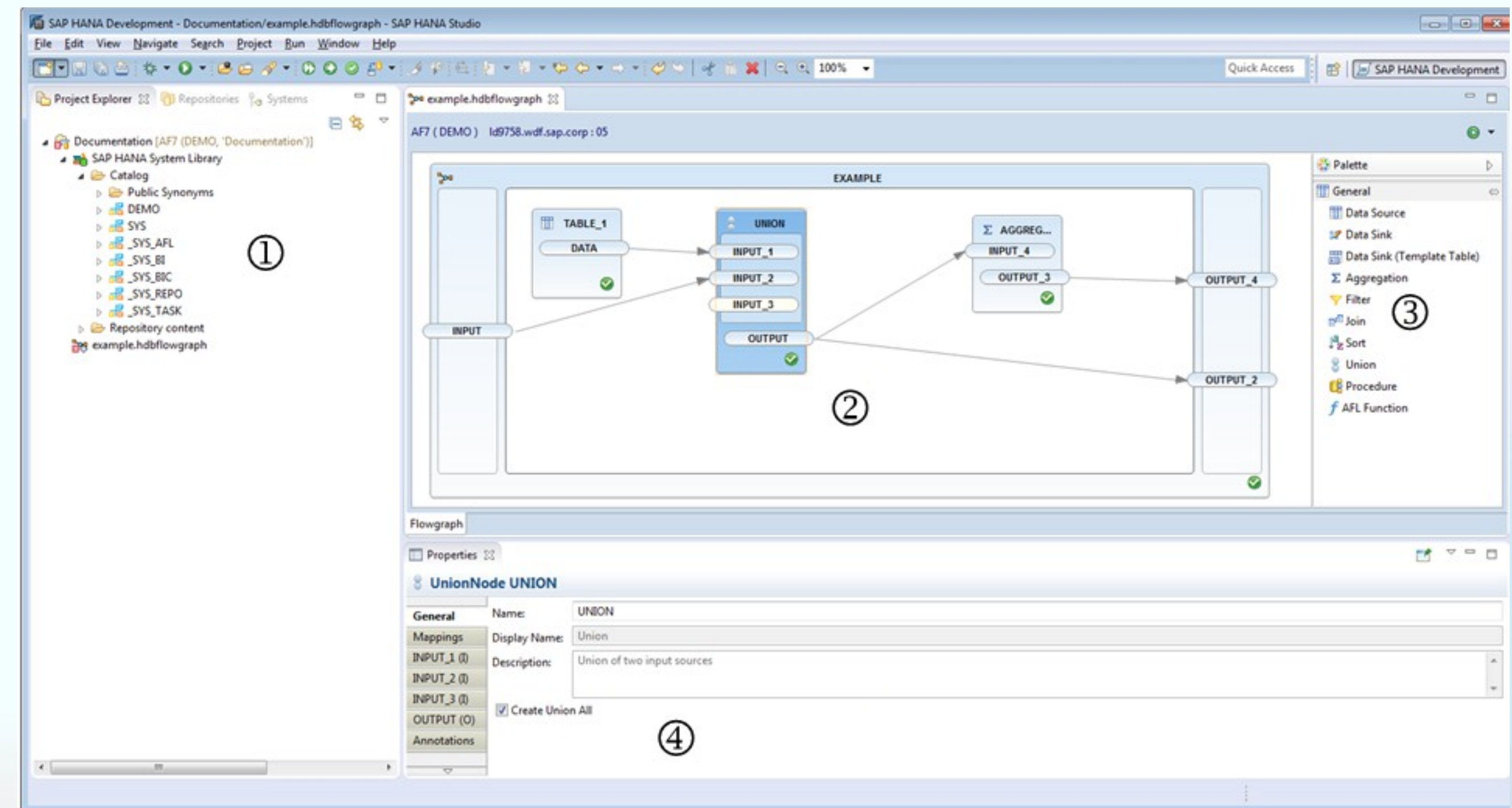
- Business Objects Acquisition 2008
- Full-featured ETL (Extract, Transform, Load) product (batch and realtime)
- Hundreds of built-in connectors (from Files to Mainframes to Snowflake...)
- Strong data-quality built in (e.g. Cleanse, Standardize, Validate)



How did it all start?

SAP HANA Application Function Modeler (AFM)

- Created ~ 2013
- Making Predictive Analysis Library (PAL) and Business Function Library (BFL) accessible through UI
- Rudimentary Support for simple Filter, Aggregation, Union, Procedure Calls

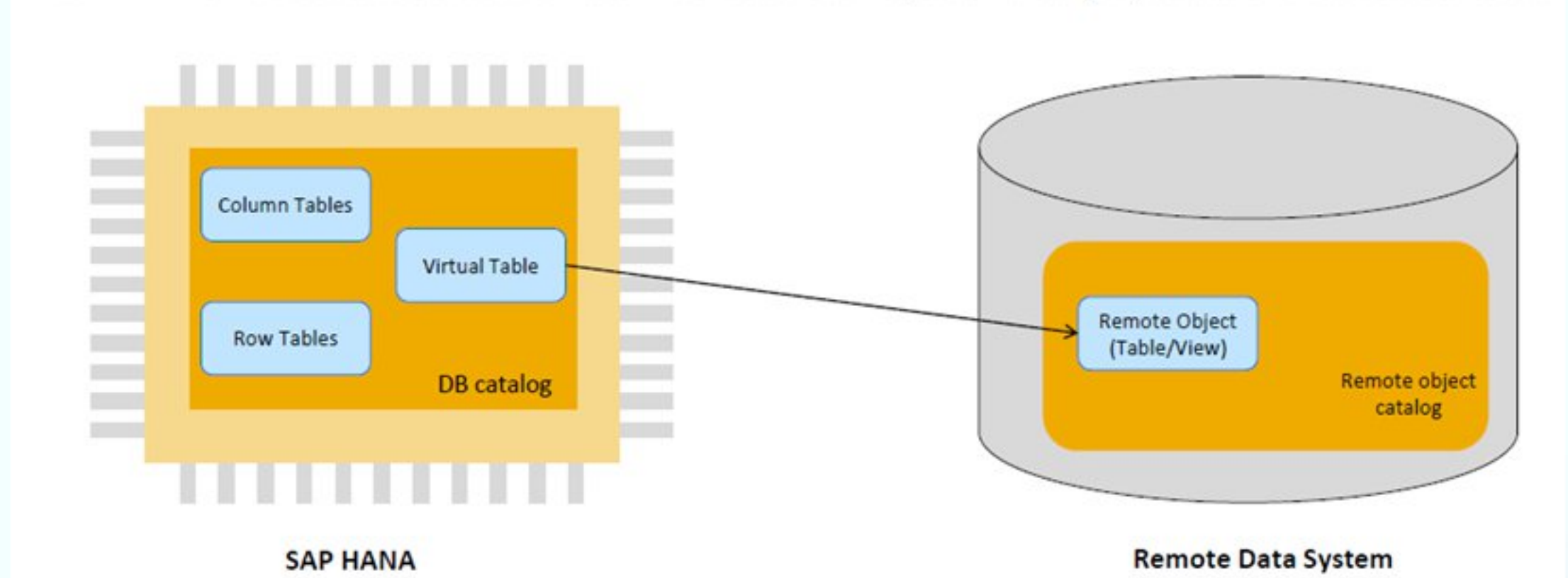


How did it all start?

SAP HANA Smart Data Access (SDA)

- Since HANA 1 SPS06
- Virtual table access (Data Federation)
(since SPS07 also insert/update/delete)
- Implemented directly in Hana Indexserver (no longer necessarily the case)
- Not extendable with custom sources besides ODBC
- Limited Query Pushdown (regarding joins, expressions and aggregations)

The SAP HANA smart data access is based on local virtual tables that maps to an existing object at the remote data source site

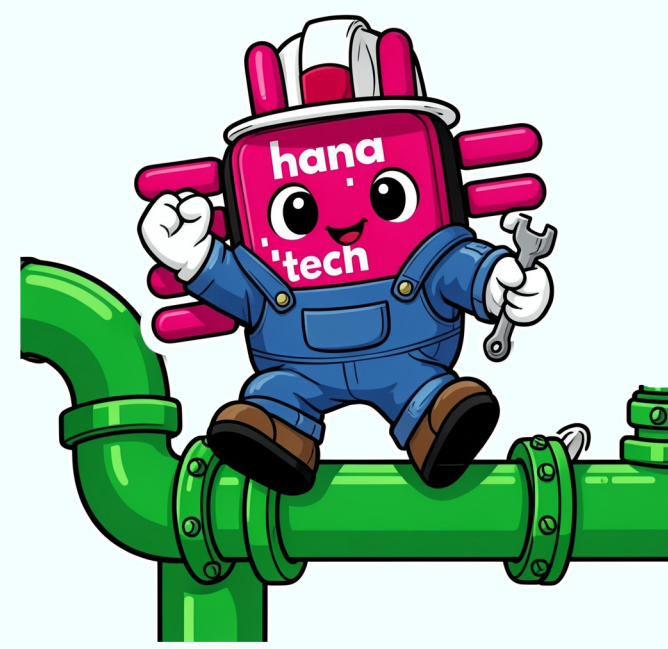


< > Sample Code

```
CREATE REMOTE SOURCE <remote_source_name> ADAPTER "hanaodbc"  
  CONFIGURATION 'Driver=libodbcHDB.so;ServerNode=<HANA_Cloud_endpoint>;  
  encrypt=TRUE;'  
  WITH CREDENTIAL TYPE 'PASSWORD' USING 'user=<user_name>;password=<password>';
```

```
CREATE VIRTUAL TABLE MYSCHEM1.HANA1_T1 AT "HANA1"."DT1"."MYSCHEMA2"."T1";
```


Smart Data Integration (SDI)

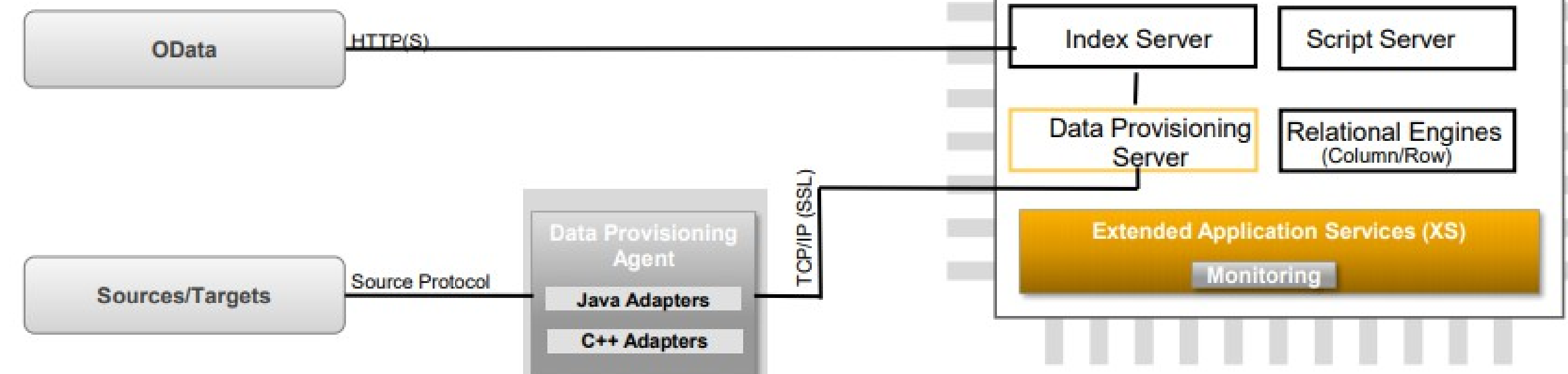


- Since HANA 1 SPS09
- DP Server and DP Agent
- Realtime (Remote Subscriptions)
- Enhanced Query Pushdown
- HDBFlowgraph & HDBReptask with Complex Data Transformations (e.g. Pivot, Table Comparison or XML flattening) and Partitioning Options

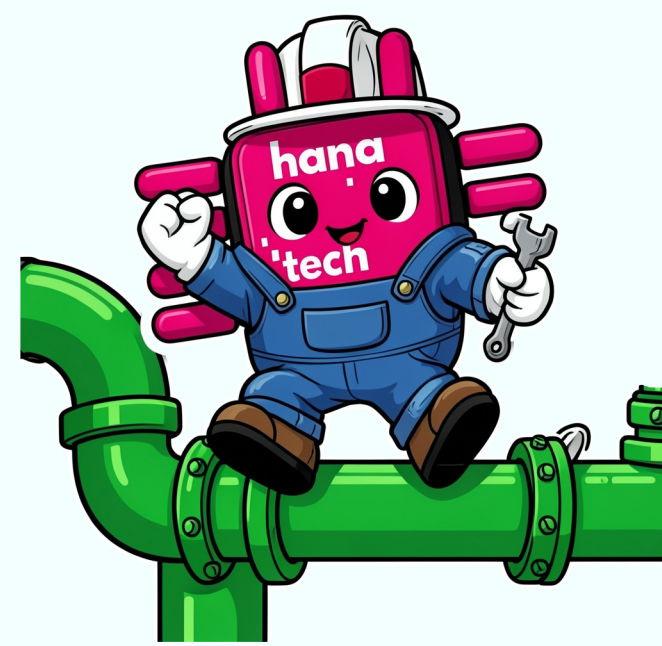
SAP HANA smart data integration Architecture – TCP/IP (SSL)

The Data Provisioning Server and Data Provisioning Agent are the two main components of SAP HANA smart data integration.

By default, the Data Provisioning Agent is configured to communicate with the Data Provisioning Server using TCP/IP.



DP Agent



- Installable on Premise
- Translates from source protocol to HANA Rows
- Cloud Connection via HTTPS
- Failover through AgentGroups
- Lots of built-in Adapters (e.g. OracleLogReader, FileAdapter, SOAP)
- Java Adapter Framework & Apache Camel Support
- Credential Management

Component	Operation System	Architecture	Version	File Size	Download
SDI Data Provisioning Agent	Linux	x86_64	2.0 SP08 Patch 12	512.2 MB	Download Link (sha1)
SDI Data Provisioning Agent	Windows	x86_64	2.0 SP08 Patch 12	536.9 MB	Download Link (sha1)

```
*****
                        SAP HANA Connection
*****
1. Connect to SAP HANA Cloud via JDBC
2. Connect to SAP Datasphere via JDBC
3. Connect to SAP HANA service for SAP BTP, Cloud Foundry environment via JDBC
4. Connect to SAP HANA service for SAP BTP, Neo environment via HTTP
5. Connect to local SAP HANA (on-premises) via JDBC
6. Connect to local SAP HANA (on-premises) via TCP
q. Quit
b. Back
*****
```


Demo

Flowgraph & Reptask

Smart Data Quality (SDQ)



- Runs in ScriptServer on SAP HANA
- Cleansing (with and without Address Directories)
- Matching & Best Record (select survivor)
- Geocode & Reverse Geocode

Demo SDQ

SDQ Cleansing (Demo Backup)



Dr. Hans Peter Schmidt
Example Incorporated
Bahnhofstrasse 1 68789 Rot
Deutschland

->

Title: Dr.

FirstName: Hans

Middle Name: Peter

Last Name: Schmidt

Example Inc

Bahnhofstr. 1

68789 St. Leon-Rot

Landkreis: Rhein-Neckar-Kreis

Regierungsbezirk: Karlsruhe

Bundesland: Baden-Württemberg

Edit Cleanse Settings

▼ Firm

Casing

☐ First Bank Corp.

☒ FIRST BANK CORP.

Diacritics

☐ Telecomunicações São Paulo

☒ Telecomunicacoes Sao Paulo

Cleanse Domain

EN_US (English - United States & Canada) ▼

Output Format

EN_US (English - United States & Canada) ▼

▼ Address

Casing

☐ 100 Main St

☒ 100 MAIN ST

Diacritics

☐ 100 Main St

☒ 100 MAIN ST

▼ Address

Casing

☐ 100 Main St

☒ 100 MAIN ST

Diacritics

☐ München Str 100

☒ Muenchner Str 100

Script Conversion

☐ Lane 81 Hutai Rd No. 34 (convert to Latin)

☒ 沪太路 81弄 34号 (preserve input script)

Street Format

Most common for each country ▼

Region Format

☐ CA

☐ California

☒ Most common for each country

Postal Format

☐ PO Box 100

☐ P.O. Box 100

☐ Post Office Box 100

☒ Most common for each country

Override Local Settings

☐ Enable suggestion list

No valid reply columns available ▼

Person Basic

☒ Prenome (Mr.)

☒ First Name (John)

☒ Middle Name (Paul)

☒ First Name and Middle Name (John Paul)

☒ Last Name (Schmidt)

☒ Last Name 2 (Miller)

☒ Last Name 1-2 (Schmidt Miller)

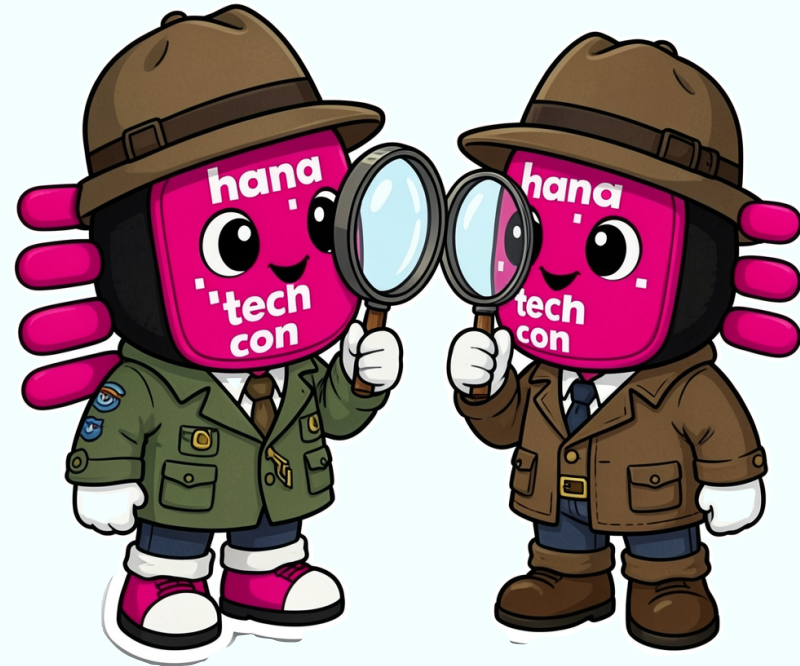
☒ Maturity Postname (Jr.)

☒ Honorary Postname (Ph.D.)

☒ Person (John Paul Schmidt Miller Jr., Ph.D.)

☒ Title (President)

SDQ Matching (Demo Backup)



▼ Address

☒ 100 Main St matches 100 Main St Suite 200

☒ 100 Main St matches 100 Main

☒ 100 Main St matches 100 Main Ave

☒ 100 Main St matches 100 N Main St

☒ 100 S Main St matches 100 N Main St

Match Strictness

Looser

Default

Tighter

Default is the best practice as it contains a good balance between identifying fuzzy matches and not many false positives

▼ Firm

☒ Royal Medical Center matches RMC

☒ Linda's Restaurant matches Linda's

☒ International Group matches Int. Grp.

☒ First Bank #72 matches First Bank #52

Match Strictness

Looser

Default

Tighter

Default is the best practice as it contains a good balance between identifying fuzzy matches and not many false positives

▼ CUSTOM2 (Custom)

☐ 123456789 matches a blank value

☐ Match if there are blank values in both records

☐ Northeast matches NE

☐ 123456789 matches 124356789 (3 and 4 are transposed)

Match strictness:

Looser

Tighter

▼ Person

☒ John Schmidt matches J. Schmidt

☒ John Schmidt matches John-Paul Schmidt

☒ John Schmidt matches W. John Schmidt

☒ John Schmidt matches Jonathan Schmidt

☒ John Schmidt matches John S.

☒ John Schmidt matches John Schmidt Bauer

☒ John Schmidt matches John Schmidt Jr.

Match Strictness

Looser

Default

Tighter

General Settings

Side effect data level

☐ None

☐ Minimal

☒ Basic

☐ Full

Tracing table output

☒ All

☐ Maximum rows

100000

Source Settings

☐ OFF

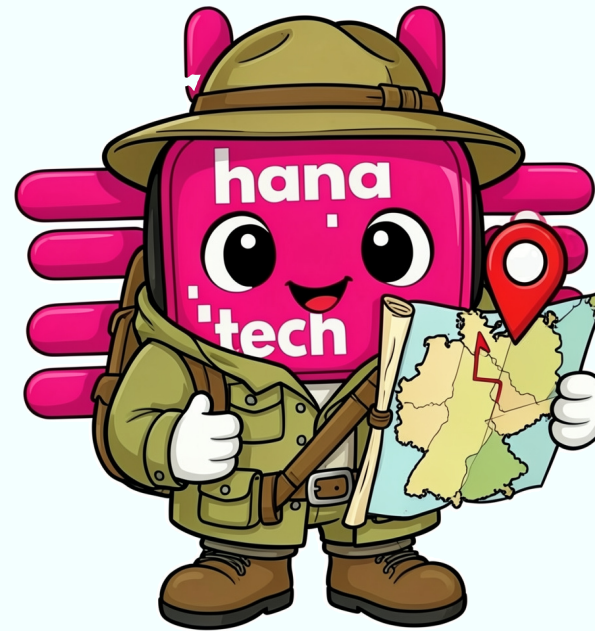
Survival Rule

The survival rule determines which record in each group of duplicates is marked as the survivor.

- **Most Recent/Oldest:** For each group of duplicate records, the record with the most recent/oldest date is selected as the survivor.
- **Priority of Sources:** Order the sources by moving the most important sources to the top. The record that contains the highest priority value in the group of duplicate records is the survivor.
- **Longest/Shortest:** Select one or more columns for this rule. For each group of duplicate records, the record with the longest/shortest value in the specified columns is selected as the survivor.

Drag the survival rules to arrange them in priority order.

SDQ GeoCode (Demo Backup)



▼ Geocode

☒ Include additional geographic and census data

☒ Include nearby addresses

Distance Unit

☒ Kilometers

☐ Miles

Radius: 5

0.0001

110

Maximum Records

Default

100

Input Column

MATCH_ADDR_BUILDING

Edit Geocode Settings

▼ Geocode

☒ Include additional geographic and census data

Assignment Level

☒ Best

☐ House Number

▼ General

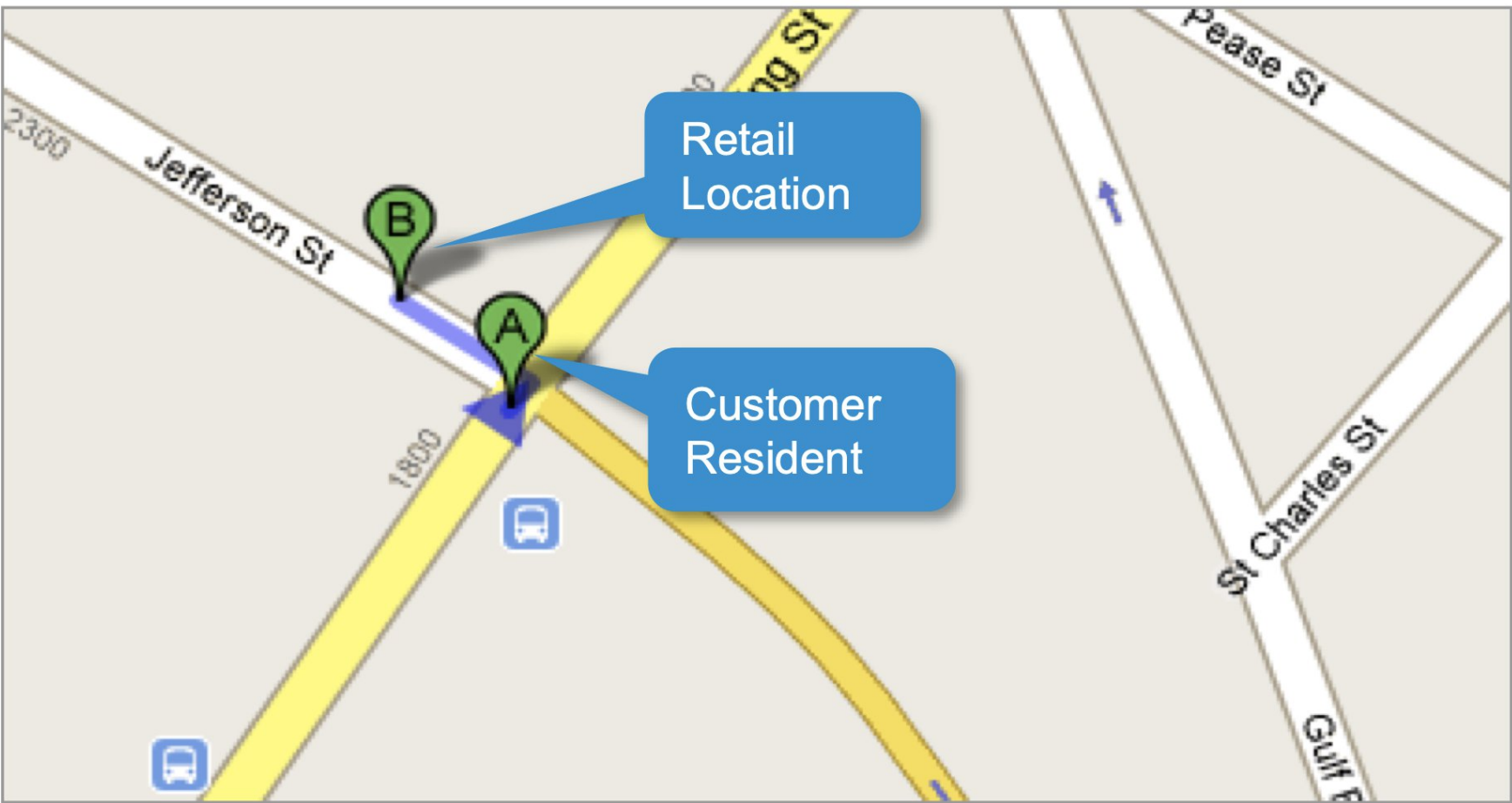
Side Effect Data Level

☒ None

☐ Minimal

☐ Basic

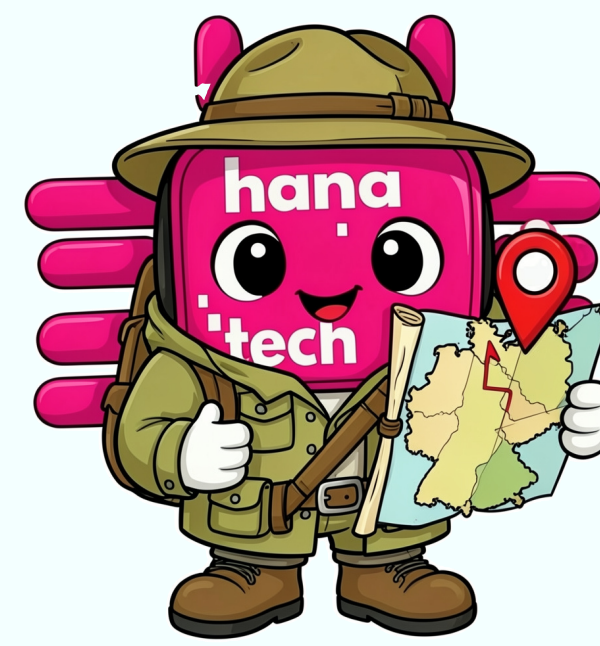
☐ Full



Example: Find the closest store for each customer

Cust Name	Customer Address	Geo Code	Closest Store #
Margaret Roberts	1801 Dowling St., Fort Collins, CO 80522	(40.575874, -105.101652)	1544
Neil Nevue	942 California Ave, Salt Lake City, UT, 84115	(40.74043, -111.935701)	4403

Related Products



SAP Data Quality Management, microservices for location data

- Standalone BTP Service with API for cleansing and (reverse) geocode
- Usable in HANA Cloud Flowgraphs
- SuccessFactors Employee Central (Address Typeahead and Address Validation)
- Integration in BDC / Datasphere through API calls in Python

Further Reading

- SDI Overview by Werner Daehn
<https://community.sap.com/t5/technology-blog-posts-by-sap/hana-smart-data-integration-overview/ba-p/13102767>
- Data Cleansing and Data Matching by Tim Huse
<https://community.sap.com/t5/technology-blog-posts-by-sap/data-cleansing-and-data-matching-with-sap-hana-smart-data-quality/ba-p/13571208>
- Getting Started with SAP Data Quality Management, microservices for location data: BTP Free Tier by Hozumi
<https://community.sap.com/t5/technology-blog-posts-by-sap/getting-started-with-sap-data-quality-management-microservices-for-location/ba-p/13527838>
- GitHub Samples for Data Quality Microservice for SAP Datasphere / BDC
<https://community.sap.com/t5/technology-blog-posts-by-sap/github-sample-for-working-with-sap-datasphere-and-sap-data-quality/ba-p/13730043>
- DQM Address Validation in CX Sales and Service Cloud V2 by Prasad
<https://community.sap.com/t5/crm-and-cx-blog-posts-by-members/enhancing-customer-data-accuracy-with-sap-dqm-address-validation-in-cx/ba-p/14313467>

Thank you for your attention :)

Questions?

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LinkedIn: [@kasimos](#)

